Hydraulic Function

General

The following description refers to the standard hydraulic system of MacGREGOR hydraulic deck cranes, GLB version.

Minor variations in the hydraulic system may occur in individual cranes of the same type, and the objective of this description is only to provide a general orientation concerning the arrangement of the hydraulic system.

Note!
For reference see hydraulic circuit diagram in section 9.4 in the parts manual. All hydraulic components mentioned in this instruction can be found in the diagram.

The crane drive system consists of three separate hydraulic work circuits, i.e. hoisting, luffing, and slewing respectively. There is also a feed and control pressure circuit, to assist the hydraulic work circuit.

Each of the work circuits may have one or more pumps located in the pump unit, in Figure 1 marked PHO, PLU, PSL. Pump PF for the feed and control pressure system is also located in the pump unit.

Identifying circuits in the hydraulic diagram

Figure 1 shows the arrangement of the three work circuits, hoisting, luffing, and slewing circuits, together with the feed and control pressure circuit. The circuits appear in the same order in the hydraulic diagram.

Symbols

Every hydraulic component is represented by a special symbol in the circuit diagram. The meaning of the symbols is explained in section "General" in the Manual. The symbol for the reservoir is shown in several places in the diagram for greater simplicity of drawing; but there is naturally only one reservoir in actual fact.

Component identification

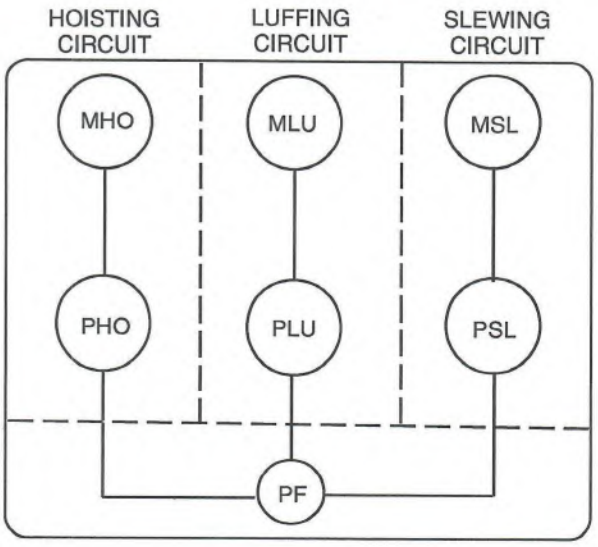
The main components are identified by four-digit numbers in the diagram and on the actual components in the crane. The first digit of each number signifies the circuit to which the component belongs:

1... Hoisting circuit
2... Luffing circuit
3... Slewing circuit
4... Feed and control pressure circuit
5... Additional programming equipment *)
6... Twin crane hydraulic equipment
7... Other additional hydraulic equipment *)

*) Equipment installed at request of customer

Legend

MHO	Hydraulic motor, hoisting circuit
MLU	Hydraulic motor, luffing circuit
MSL	Hydraulic motor, slewing circuit
PHO	Pump, hoisting circuit
PLU	Pump, luffing circuit
PSL	Pump, slewing circuit
PF	Pump, feed circuit



FEED AND CONTROL CIRCUIT

Fig. 1. Identifying hydraulic circuits.

Feed and control circuits

Feed pressure circuit

The function of this circuit is mainly to replace the hot fluid leaking from the closed work circuits. The feed fluid is introduced through non-return valves into the low-pressure side of the system. The feed circuit operates at a pressure (inlet/outlet) of approx. 3.0/ 2.5 MPa (30/25 bar), which is set at valve 4148 and is measured at test point 1 and 16 respectively. Test point 1 is feed pressure inlet and 16 is feed pressure outlet.

Control pressure circuit

The function of this circuit is to supply fluid to the brakes and to the pump servos, that controls winch speed, slewing speed and working direction. The system operates at the same pressure as the feed pressure and is measured at test point 1.

Note!

For exact pressure settings, see pressure list in the hydraulic diagram.

Hoisting winch circuit

Introduction

The hydrualic circuit diagram shows the following main hoisting components:

Pumps	1111, 1112
Hydraulic motor	1141
Brake	1211

The brake is mounted inside the winch gearbox.

Function

With the control lever in neutral, Figure 2 or 5, and the pump running, the hoisting circuit performs no actual work. A continuous supply of fluid from the feed circuit, flushes the circuit.

Low-speed range, brake release

When the control lever is moved to the left for low-speed operation of the winch, Figure 3, valve 1221 and 1226 in the actuated mode, "b", (1226 slightly delayed in relation to 1221) and connects full control pressure to the brake valve 1226 which releases the brake, and also to valve 1128-1 which shifts from its unloading mode.

With no load on the cargo hook and the control lever in brake release position, Figure 3, the hoisting winch should produce a slow hoisting movement, so called plussing. See instruction "High pressure pumps" in section 6.1.

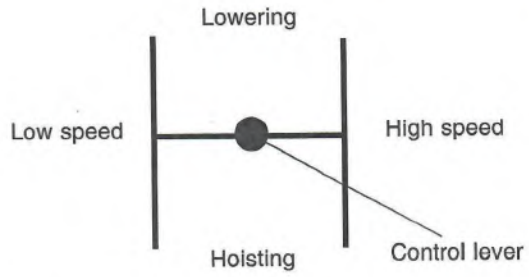


Fig. 2. Control lever in neutral, hoisting

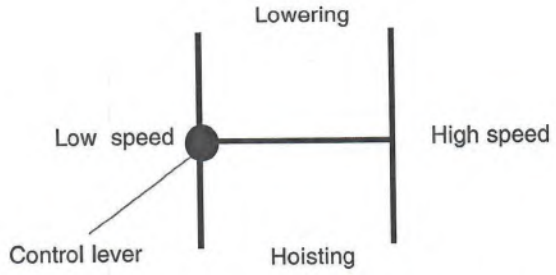


Fig. 3. Control lever, hoisting, brake release position

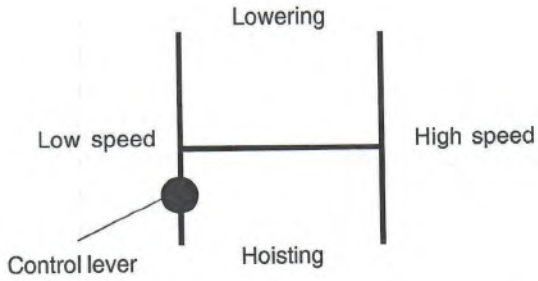


Fig. 4. Control lever, hoisting, low-speed hoisting

High speed range, brake release

When the control lever is moved to the right, Figure 6, valves 1221 (a) and 1226 (b) are actuated, (1226 slightly delayed in relation to 1221). Control pressure is connected to the brake valve 1226 which releases the brake, and also to valve 1128-1 which shifts from its unloading mode.

The brakes are released and the motor assumes the high-speed position (min. displacement). The motor rotates at double speed, and the lifting capacity of the winch is reduced to approximately 50 - 60 % of its maximum rate.

With no load on the cargo hook and the control lever in brake release position, Figure 6, the hoisting winch should produce a slow hoisting movement in the same way as in low speed operation.

High-speed range overload

If the load on the cargo hook is too heavy for handling in the high-speed range, the pressure switch 1311 will cause valve 1221 to shift to low-speed range. Full displacement on the hydraulic motor.

Hoisting and lowering, high speed or low speed

Pulling the control lever backwards from the brake release position will start the winch to lift the load, Figure 4 low speed and Figure 7 high speed. The servo valves 1111-6 ,1111-7, 1112-6 and 1112-7 affect the servo cylinders and thus the pump displacement. The pumps 1111 and 1112 thereby produce a flow of fluid that is proportional to deflection of the control lever.

If, instead, the control lever is moved forward, the load will be lowered, ie, the function is the same as for hoisting, except for the direction of flow.

High speed lowering (optional)

If the crane is designed for high speed lowering the motor will automatically assume high-speed position when the control lever is moved forward for lowering, even if the control lever is to the left, for low speed. Valve 1221 (a) will be actuated.

Safety functions

The hoisting winch circuit embodies three safety features:

- Main pressure limiting valves 1111-2; 1111-3, 1112-2, 1112-3.
- Pressure switch 1311 (overload valve) stops the crane in the event of overloading at low speed. Shifts to low-speed when running in the high-speed mode.
- Pressure switch 1381 stops the crane in the event of insufficient feed pressure.

Note! It is not allowed to adjust sealed safety valves without contacting MacGREGOR Cranes, Service Department.

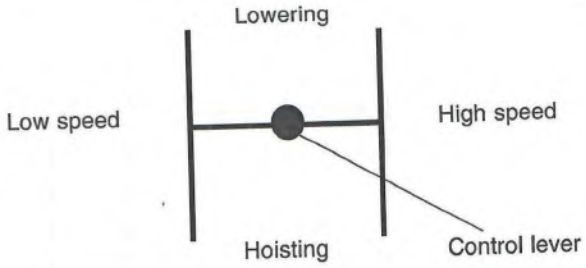


Fig. 5. Control lever in neutral, hoisting.

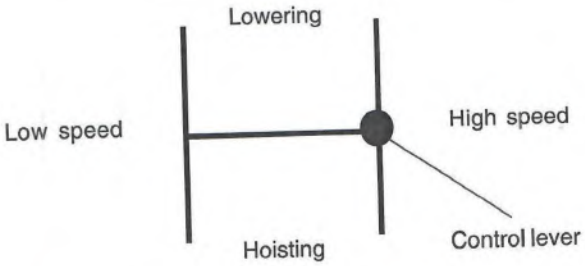


Fig. 6. Control lever, hoisting, brake release position.

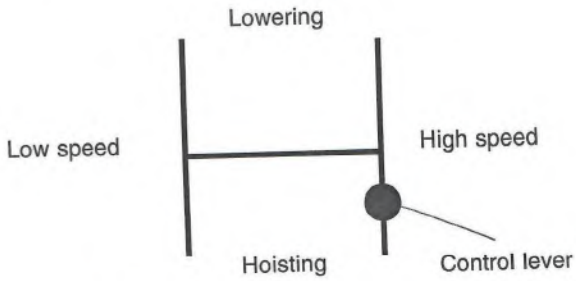


Fig. 7. Control lever, hoisting, high-speed hoisting.

Luffing winch circuit

Introduction

The hydrualic circuit diagram shows the following main luffing components:

Variable displacement pump	2111
Hydraulic motor	2141
Brake	2211

The luffing and slewing movements are generally operated by a common control lever, Figure 8.

Note!
It is not allowed to adjust sealed safety valves without contacting MacGREGOR Cranes, Service Department.

Fig. 8. Control lever, luffing/slewing.

Brake release

Moving the control lever from neutral, eg. backwards to raise the jib, causes the valves 2221 and 2226 (2226 slightly delayed in relation to 2221) to shift position and control pressure is admitted to the brake 2211 and to valve 2127 which shifts position from the unloading mode.

With the control lever in the brake release position, the luffing winch should produce a slow movement of the crane jib, so called plussing. See instruction "High pressure pumps" in section 6.1.

Raising and lowering the jib

Moving the control lever further backwards starts operation of the luffing winch. The servo valves 2111-6 and 2111-7 affect the servo cylinder and thus the pump displacement. The pump 2111 thereby produces a flow of fluid at a rate that is proportional to the lever deflection.

To lower the jib, move the lever forward from neutral. The function is the same as for luffing-in, except for the direction of the fluid flow.

Safety functions

The luffing winch circuit embodies two safety features:

- Pressure limiting valves 2111-2, 2111-3.
- Pressure switch 2481; stops the crane in the event of insufficient feed pressure.

Slewing circuit

Introduction

The hydrualic circuit diagram shows the following main slewing components:

Variable displacement pump	3111
Hydraulic motors	3141, 3142
Flush unloading unit	3127
Brakes	3211, 3212

The luffing and slewing movements are generally operated by a common control lever, Figure 9.

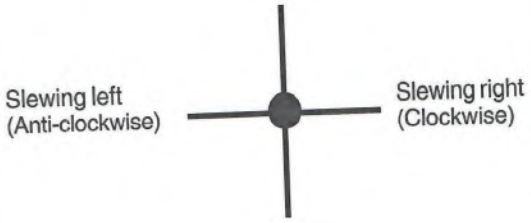


Fig. 9. Control lever, luffing/slewing.

Brake release

Moving the control lever, eg. to the left for slewing left (anti-clockwise), will affect valve 3221. The brakes will be released and the unloading valve 3127-1 will be blocked.

Slewing left and right

By moving the control lever further to the side, the crane will start to slew in the direction chosen. The servo valves 3111-6 and 3111-7 affect the servo cylinder and thus the pump displacement. The pump 3111 thereby produces a flow of fluid at a rate that is proportional to lever deflection.

To slew to the right (clockwise), move the lever to the right instead. The function is the same as for slewing left, except for the direction of the fluid flow.

Safety functions

The slewing circuit embodies one safety feature:

- Main pressure limiting valves 3111-2 and 3111-3.

Starting and Stopping, Electric Safety Functions Single Cranes

Introduction

The following description refers to the starting and stopping, and electric safety functions, of a standard-type single crane. A complete electric circuit diagram relating specifically to the particular crane will be found in the section SPARE PARTS of the Manual.

The purpose of the present description is to furnish a general idea of the design and function of the electric system. The crane is supplied with electric power via the slip-ring unit and the main switch in the crane house. Cables lead from the main switch to: a transformer supplying power for all control circuits; a switch for the oil cooler fan motor; a Y/D switch for the main electric motor; and a thermostat switch for the ventilation fan in crane house.

All components are marked by symbols in the diagram, making it easier to follow the various stages of the starting sequence, e.g.

- Contactors: K1 (N); K3 (D); K2 (Y).
- Time relays: K9; K71.

Before start

Verify that the ambient temperature is not below -15°C. If the crane is operated in temperatures below -15°C a heater has to be installed in the oil tank.

Starting and stopping the crane

Note! Joysticks for hoisting or luffing/slewing motions must be in neutral position when starting the crane. If the joysticks have been moved in any direction when pressing the start button the motion is blocked. Move the joysticks to neutral position then operation is possible.

Note! If crane is equipped with PMS a starting permission is required. Read "OPERATING INSTRUCTION START - STOP" and "Controls and Indicators" to operate crane with PMS.

Start sequence

- Press button S13 "1" (start).
- Pilot light H1 "DO NOT OPERATE" goes on.

The feed circuit pressure build up its operating pressure. Pressostats BP1 - hoisting circuit, BP2 - luffing circuit, close and start a delayed closing 8s time relay K71.

- After 8s, Y contactor K2 switches on N contactor K1; the main motor M1 starts in Y connection.
- At the same time as the Y contactor was switched on, an 8s delayed-opening double relay K9 was started.
- After 8s, the Y contactor drops, and D contactor K3 is switched on by double time relay K9 closing after 0.1s.
  The D contactor and self-holding N contactor switch main motor M1 to delta configuration.
- Push button "1" (start) closing prepares the energizing of motor switch Q5 of oil COOL FAN MOTOR M3.
  Motor M3 is started by thermostat BLT1 if the temperature of the hydraulic oil exceeds +10°C.
- Warning light H1 goes out. The crane is operational.

Stop sequence

- Press button S13 "0" (stop).
- All contactors of Y/D switches and all other motor contactors are de-energized. All motors stop.
- The ventilation fan motor stops only when the main switch Q1 is switched OFF.

Other electric functions

Overheating protection

A thermostat BT2 in the hydraulic system of the crane opens the control circuits of the main electric motor and the oil cooler fan motor when the oil temperature exceeds +85°C. Pilot light (H5) "High oil temperature" illuminates.

Restart the crane by use of pushbutton S13 "1/0" (start/stop). Pilot light H1 "DO NOT OPERATE" illuminates. Crane and cooler fans start, but the crane control system CC3000 is still blocked.

Pilot light H5 goes out when the crane's working oil temperature is back below +85°C. The crane control system CC3000 returns to normal and pilot light H1 goes out.

Oil level switch

If the oil level in the hydraulic oil tank drops below the minimum permissible level, an alarm system is switched on by a float type oil level switch BLT1. Warning light H2 goes on.

If the oil level sinks further, level switch BLT1 opens the control circuit of main motor M1.

Siren

A pushbutton "SIREN" in the cabin allows the operator to warn people on deck.

Ventilation fan

For all cranes a ventilation fan is fitted in the crane house. This is controlled by a +25°C thermostat BT1 which is located in the CT1-cabinet.

Lighting and heating

Electric power is supplied by separate slip-rings for lighting in the crane house, in the operator's cabin, and on the jib and heating in the cabin.

The crane house lighting is operated by a switch under the bottom plate. The cabin lighting is operated by a switch and the jib light is operated by a pushbutton in the cabin.

Control system error

If an error is detected by the crane control system in the CT2-cabinet pilot light H4 illuminates. Check the CT2-cabinet according to instruction "Crane Control System CC3000" under Section 6.3.

Crane Control System CC3000

General

The following description refers to the microcomputer based crane control system CC3000 for MacGREGOR Cranes hydraulic deck cranes, single crane versions.

Minor variations in the use of control system input and output signals due to differences in hydraulic and electric system may occur. The objective of this description is only to provide a general orientation concerning the principles of the control system.

For detailed information about the control system a complete electric circuit diagram, relating specifically to your crane, will be found in the instruction manual, section "Spare Parts", Group 9.5. Also for detailed information about the hydraulic system a complete hydraulic circuit diagram can be found in section "Spare Parts", Group 9.4.

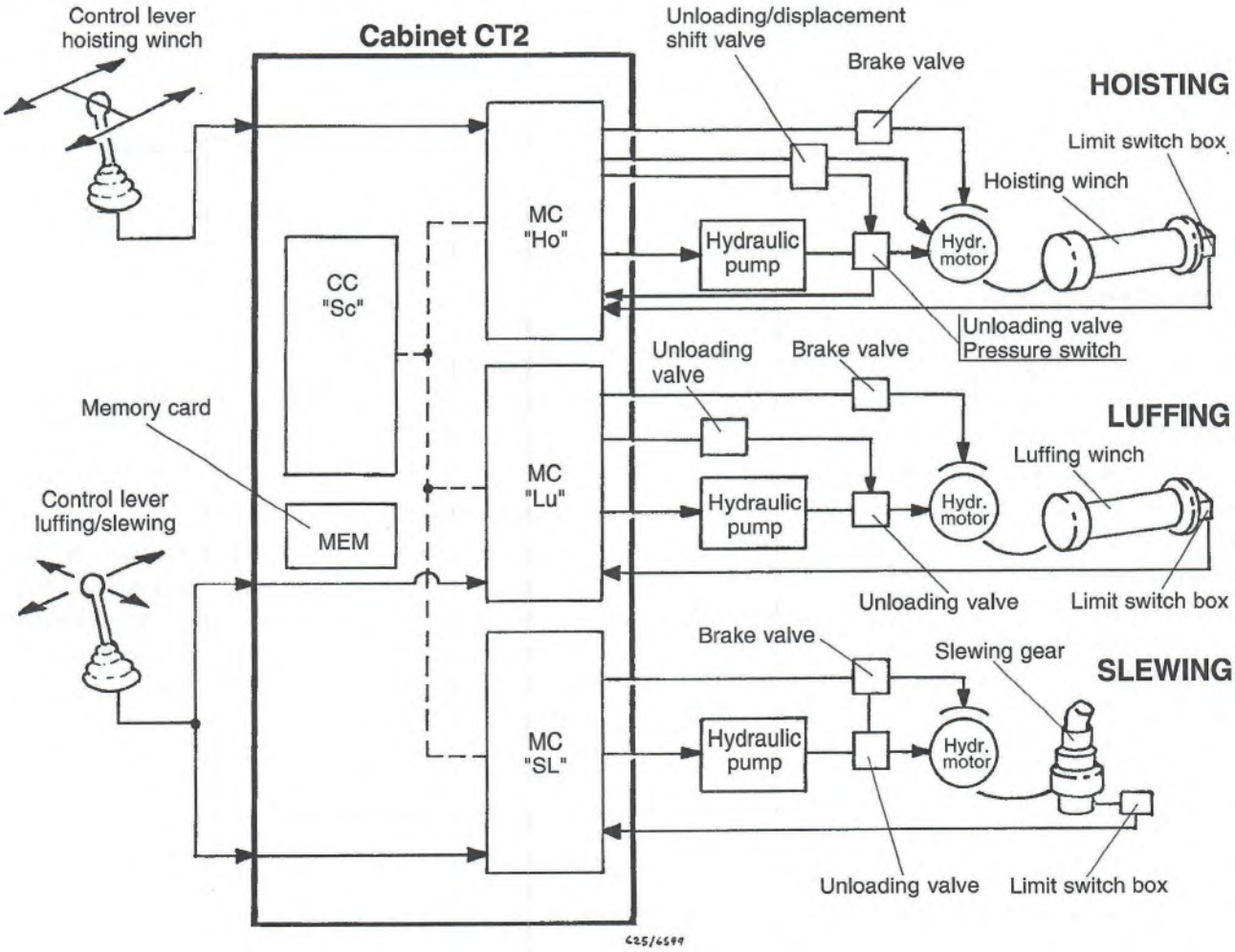


Fig. 1. Crane control system with interfaces to control levers and hydraulic system.

Crane control system - interfaces and principle of operation

Control levers (joysticks)

The right control lever controls the hoisting winch. The signals from the control lever are connected to the MC card, "Ho" for hoisting winch.
The left joystick controls the luffing and slewing movements. Both digital and analouge signals are connected to MC card "Lu" for luffing and "SL" for slewing.

The joystick linear potentiometers are fed with 0V, +6V, and +12V. The output level is approximately +6V with the joystick in neutral. The output signal is controlled by the joystick and is decreased to approximately +1.5V and increased to approximately +10.5V, depending on the joystick movement.

The joystick for hoisting also gives 24V signals for selection of low speed/high speed and hoisting/lowering. The joystick for luffing/slewing also gives 24V signals for selecting luffing in/luffing out and slewing left/slewing right.

Crane control system

The Crane control system consists mainly of four micro processor cards. One Communication Card (CC) is used for internal communication between the cards and a parameter memory card. Three Motion Control cards (MC) are controlling all input and output signals for each crane movement hoisting, luffing and slewing. Each MC card feeds a potentiometer in the control lever for the crane movement in control. Potentiometer and control lever direction signals go to a microprocessor with a software control program that handles brake release logic, overload logic and rampforming. The output from the microprocessor is fed to a voltage to current amplifier on the card.

The rampformer gives smooth acceleration and retardation to the crane movements.

The brake release logic circuit releases and applies the brake in the correct sequence with the displacement/unloading signals when the hydraulic motor is started and stopped.

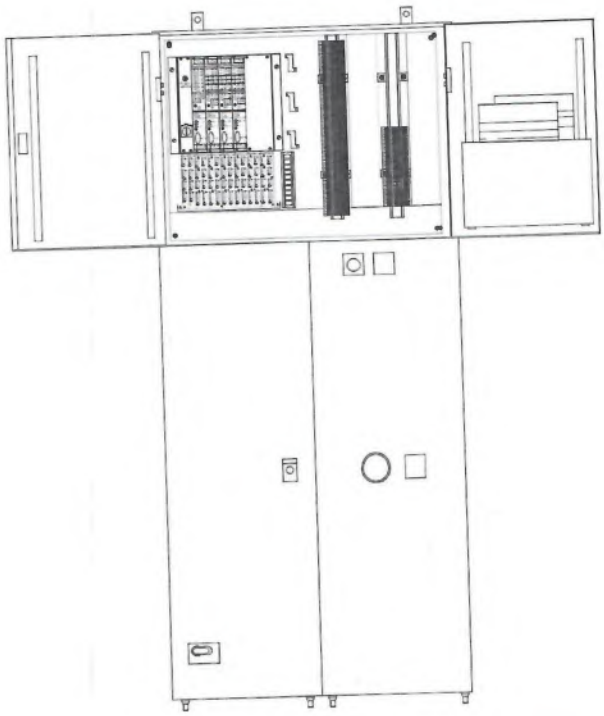


Fig. 2. Crane control system CC3000 in CT2 cabinet.

The voltage to current amplifier feeds the hydraulic pump solenoids.

CT2 cabinet

CT2-cabinet containing the crane control system is protected against humidity by a casing with protection class IP54 (dust and splash water proof).

19" subrack with electronic cards

All electronic cards are mounted in a standard 19" aluminium subrack. A backplane card is used for connecting power, 24V to the CC and MC cards. It also provides communication lines between CC and MC cards and between CC card and the parameter memory card.

Hydraulic pumps

The hydraulic pumps are variable displacement pumps (e.g., the flow is steplessly variable from zero to max. rated flow). The pump oil flow is controlled by a 24V pulse width modulated (PWM) current signal. The direction of the oil flow is dependent on which one of the two solenoid valves is activated.

The speed of the winch, the pump displacement, the current through the solenoid coil and the lever movement are proportional to each other.

Hydraulic motors

The hydraulic motors may be reversed by changing the direction of the hydraulic pump oil flow. Each motor also having a low and a high speed range selected by the operator's control lever.

Overload protection

To prevent the crane from being overloaded, i.e. lifting a too heavy load, the hydraulic system is equipped with an electro-hydraulic overload system. Pressure sensing switches, which normally enable the operation, stop the hoisting movement if a too heavy load is lifted at the current jib outreach. The pressure switch gives a signal to the control system overload logic, which senses the overload input signal and switches over from high speed to low speed or from low speed to stop.

Limit switch box, hoisting winch

The limit switch box contains cams with limit switches and encoders. It controls the movements of the hoisting block.

Limit switch box, luffing winch

The limit switch box contains cams, limit switches and encoders. It controls the movements and outreaches of the jib.

Limit switches, slackwire

The limit switch for hoisting slackwire stops the lowering movement, and the limit switch luffing slackwire stops the luffing out movement when a wire is slack.

Luffing / slewing limits

If installed, luffing/slewing limits prevent collisions between the jib and other cranes, or other objects on the vessel. For more information about luffing/slewing limits see section 4 and 6.3.

Micro Processor Cards - CC and MC

Hardware

The control system consists mainly of one CC card and three MC cards. All MC cards are identical and interchangeable.

Each MC card is a complete stand-alone microprocessor system with processors, several types of memories and communication channels, analogue and digital inputs as well as analogue and digital output current drivers. Each card can fully handle all input and output signals needed for one hydraulic circuit, i.e. one crane movement.

A safety system consists of a secondary micro processor, which monitors the behavior of the primary processor in critical situations such as hazardous operation at low feed pressure and overload pressure. This system will stop the crane movement due to activated limit.

Software

All cards, CC and MC are programmed with the same crane control software. Each card senses which rack-position and therefore which function/crane movement it is controlling.

All cards identify themselves by showing a two-character abbreviation of the function/movement name on the front panel display:

- Sc - supervisor
- Ho - hoisting
- Lu - luffing
- Sl - slewing.

The leftmost card, the CC "Sc", is only used for supervisory functions such as communication with other cards and distribution of control system parameters from the parameter memory card.

New versions of the crane control software can be loaded/programmed into the cards by using a standard notebook PC.

Parameter memory card

All parameter values are stored in the parameter memory card. The information is stored in electronic memories on the card. Stored parameter values are not dependent on power supply or battery. There is, however, a battery on the card used for a calendar, giving all cards date and time information. The card is located inside the rack and is mounted on the backplane card that also holds all the card connectors. The lifetime of the battery is 10 years.

Anti-condensation heating

To avoid condensation on the electronic cards, the temperature inside the cabinet is more then the temperature outside. Heating elements are therefore mounted behind the row of plinths. They are powered with 24V AC when the control system is switched off or when the crane is not operational.

Signal overview CC3000

Coloured LEDs

Signals to and from each card are indicated by green, yellow and red LEDs (Light-Emitting Diodes) on the front panel of each card.

Green LEDs are used for normal input and outputs.
Yellow LEDs indicate that the crane motion concerned is driven with slower speed.
Red LEDs indicate that the motion is stopped.

Signal overview sign

All signals indicated with LEDs are listed on the SIGNAL OVERVIEW sign inside the cabinet, see section "Service", Group 6.3.

Error messages

Built-in diagnostics system

The CC3000 control system has its own built-in diagnostics system. The system is capable of detecting several types of malfunctions.

The system checks itself and external input and output signals. Internal problems such as loss of communications are continuously checked. Loose connections to pumps and valves are detected when the output is activated. Signal levels that do not correspond to normal levels, i. e. the potentiometer input from the operator's handle, are also detected.

When any type of malfunction is detected, the card flashes with its error LED, no. 28. All card error LEDs are wired together to an error lamp in the operator's cabin, the "control system error" lamp. To get the corresponding error message number, push the Select/Info push-button on the card with the flashing LED.

If there are several errors, they are shown one at a time for each push on the Select/Info push-button. When there are no more error messages to read, the display shows two hyphens "- -". The error LED is now either on or off.

- If it is on, the error is still present but has been acknowledged.

- If it is off, the errors are no longer present.
When a new error is detected the error LED starts flashing again to indicate that a new error exists.

Error message sign

The corresponding error message text can be found on the ERROR MESSAGES sign in the cabinet, see section "Service", Group 6.3.

Thermostatic Controlled Ventilation Fan

General

The ventilation fan is thermostatic controlled and situated inside the crane house top, see Fig. 1.

Function

With the main switch in position "1" (on-position) the ventialtion fan starts via a thermostat, positioned in the CT1-cabinet, see Fig. 2. The thermostat bulb is situated outside the cabinet.

The thermostat is activated at +25°C. The thermostat can be adjusted, see Fig. 3.

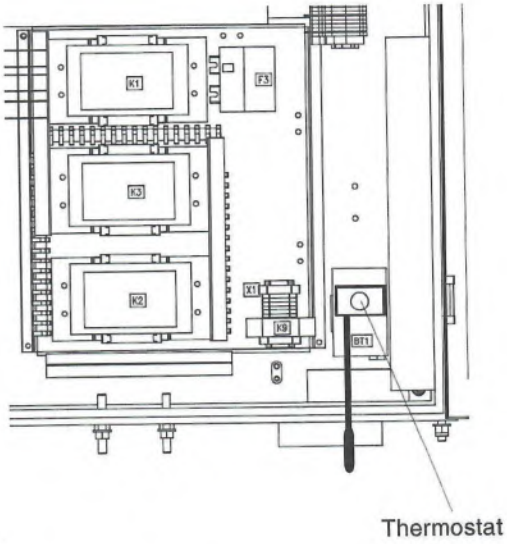


Fig. 2. Thermostat installed in CT1-cabinet

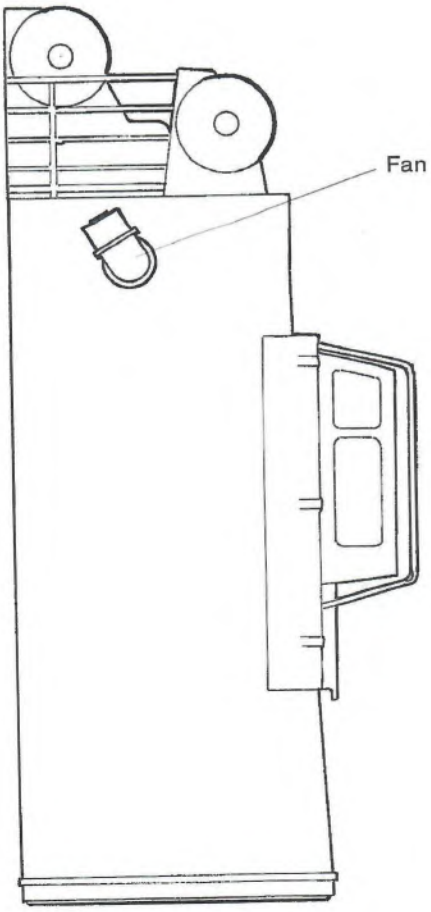


Fig. 1. Crane house

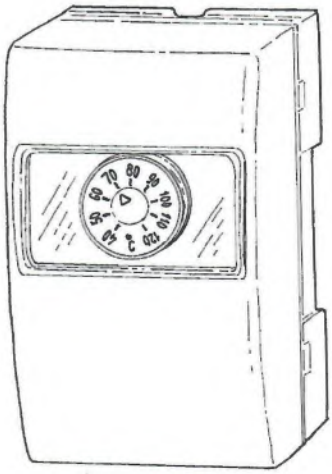


Fig. 3. Thermostat

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Tilt Warning System to Prevent "Jack Knife"

General

The tilt warning system is a safety function to prevent "jack knife", i.e. the jib and load falls into the crane house. "Jack knife" can occur if:

- The crane house tilts rearward
- The jib is in a short outreach position
- Wind blows at high speed toward the front of the crane. See Fig. 1.

The tilt warning system has a tilt switch on the jib bearing flange to sense the tilt of the crane house. The outreach for tilt warning system is sensed by either a safety cam in the the luffing winch limit box or by a software limit in the crane control system.

Function

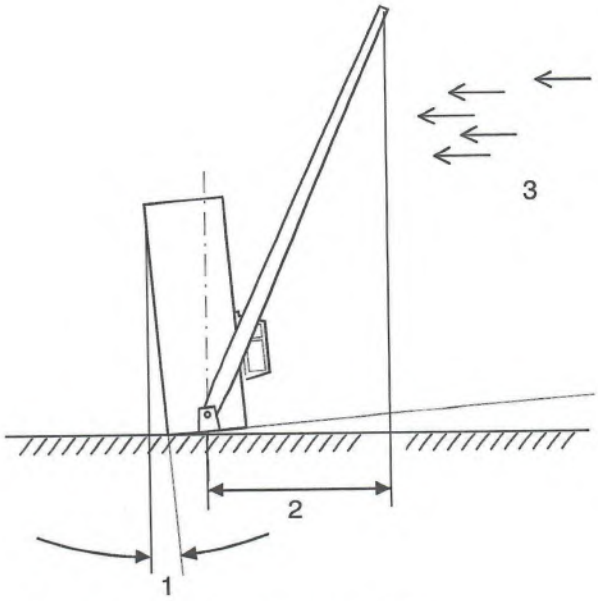
When the crane house tilts rearward more than 3° (or 3.5°), see Fig. 1, pos. 1, and the jib is in the extra minimun "jack knife" outreach, see Fig. 1, pos. 2, the luffing-in motion gets blocked. The "Tilt Warning" pilot lamp, see Fig. 2, comes on. The crane must be luffed-out and/or slewed until the "Tilt Warning" pilot lamp goes off to luff in again.

The "jack knife" outreach is noted in the instruction "Technical Data" in section 2 in the Instruction and Parts Manual.

To set the safety cam values, read instruction "Technical Data".

The tilt switch is calibrated and does not need to be set.

Read also instruction "Controls & Indicators" in section 4 in the Instruction and Parts Manual for more information.



1. Crane house tilts rearward, 3° (or 3.5°)
2. Short jib outreach
3. High wind speed toward the front of the crane

Fig. 1. Causes of "jack knife".

Note!
The Pilot lamp on the control panel are optional.

Fig. 2. "Tilt Warning" pilot lamp in cabin.

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